

Akshar Ravichandran

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Education

Georgia Institute of Technology, College of Computing

May 2026

Bachelor of Science in Computer Science, Artificial Intelligence & Devices (GPA: 3.94)

Atlanta, GA

Relevant Coursework: Machine Learning, Systems and Networking, Artificial Intelligence, Robotics and Perception, Algorithm Design & Analysis, Computer Networking, Data Structures & Algorithms, Object-Oriented Principles

Programming: Python, Java, C, C++, SQL, JavaScript, React

Tools/Libraries: React, NumPy, PyTorch, Pandas, Polars, OpenAI, AWS, Docker

Experience

Amazon

May 2025 – Aug 2025

Software Developing Engineer Intern

Tempe, AZ

- Developed one stop shop internal portal using React, JavaScript, and Amazon's Harmony Framework, streamlining certification management for 20+ PMs and SDEs supporting Amazon Business sellers
- Implemented 5 key functions, including Auditing Certifications in various regional marketplaces, Creating/Editing Certification Types, and updating 3 of our teams Coral API services, reducing manual workload by 30%
- Established and deployed 2 production links connecting my services to both production and corporation environments enabling end to end visibility for multiple stages of the portal

Sparksoft Inc.

June 2024 – Nov 2024

Full-Stack Software Engineering Intern

Columbia, MD

- Participated in developing the Enterprise Portal for Medicare under the Centers for Medicare & Medicaid Services (CMS), specializing in maintaining and refactoring the Angular-based portal
- Diagnosed and resolved 40+ front/backend bugs and performance issues, enhancing user experience of the CMS Enterprise portal

Research

Pavement Condition & Deterioration Forecasting

Sep 2025 – Present

Prediction Pipeline

Atlanta, GA

- Developed a python-based forecasting pipeline to predict pavement deterioration (MRI/IRI) using 40,000+ roadway, traffic, climate, and structural features by merging 5+ federal and municipal datasets
- Implemented a combination of Ridge, Lasso, PCA-linear regression, Random Forest, and XGBoost models to achieve total of 0.065 RMSE in next year MRI prediction, leading to an improvement in accuracy of 30% compared to a baseline regression model
- Built a recursive multi-year deterioration simulator to identify road segments at risk and align with repaving cycles, aiming to enable a proactive system to reduce the number of potholes related accidents

Machine Learning for Financial Markets VIP Research

Jan 2024 – Present

Monetary Policy Analysis (NeurIPS 2025 Dataset & Benchmarks Track – Accepted Paper) 

Atlanta, GA

- Co-authored NeurIPS accepted paper which created World Central Bank Dataset, the most comprehensive monetary policy corpus
- Trained and benchmarked Pretrained and Large Language Models to accurately distinguish policy sentiments, enhancing classification accuracy and providing deeper insights into monetary policy trends in over 25 countries, running 15,075 experiments
- Annotated and reviewed 1,000+ sentences classifying stance, temporal classification, and uncertainty; authored 2 comprehensive annotation guides to standardize the labeling

Projects

Investments Committee of Georgia Tech (IC)

Feb 2024 – Present

Quantitative Analysis Senior Analyst

Atlanta, GA

- Developed a state-space model with Kalman filter updates to dynamically estimate an equity's beta daily, to enable hedging strategies and improving backtested returns by 12% across multiple different market windows
- Implemented an automated live trading system using Interactive Brokers (IBKR) and Trader Workstation executing real-time trades, achieving a 20% reduction in transaction latency

Athletics Project Data Science Club @Georgia Tech

Sep 2023 – May 2024

Athletics Project Developer

Atlanta, GA

- Collaborated with group of 20+ to develop an accurate win probability model for NCAA College Basketball (CBB) teams using advanced Monte Carlo simulations, improving prediction accuracy by 25%
- Developed a web scraper to retrieve college basketball team statistics from 500+ games, improving the data collection process

Additional Information

Languages: English (Native), Spanish(Advanced), Tamil (Advanced)

Organizations: Georgia Tech: Startup Exchange, Pi Kappa Alpha Fraternity (Alpha Delta Chapter)